**From:** **Emanuele Mazzola** mazzola@ds.dfci.harvard.edu
**Subject:** Re: Model code for CT/GC incidence & prevalence paper
**Date:** October 1, 2021 at 11:10 AM
**To:** Roenn, Minttu Minna-Maarit mronn@hsph.harvard.edu

EM

OK Minttu, here it is attached. I think a few explanations are needed.

0) The codes are divided into a) Codes for cleaning the excel files containing the raw data I got from Meghan and preparing the input datafile for the rjags model; b) the model code itself; c) codes for plotting.
Specifically, the codes for plotting generate a) my usual/multi-paneled plot you may remember, using standard R plots (**newplots_Nov19.R**) and b) the plots and tables for the paper using ggplot (**plotsJosh_Nov19_comment.R**).

1) **Data prep** The folder contains 4 files. The "generate_CT/GC_data" should take the excel files and put them into a complete and readable format for R. "data_prep_for_mcmc.R" adds some missing fata from the previous ones
and deals with missing standard deviations for instance, preparing everything into an array useful for the rjags input.
"ProjectedDataPopulJosh.R" contains the population data for projections from a .csv file downloaded I think from the NHANES website.
These are generally only data cleaning and manipulation.

2) **Model code**
LInes 1-177 are for importing the data and adding the parameters I need in the model. They should be pretty much be explained in the code itself. Note that this is a "last version" after MANY, MANY iterations, so
I had finally decided to use the data into arrays with indexes
j= for the years
k=for the number of age groups (3)
g=for the genders
d= for the number of diseases (2)

Also within the jags model code (lines 178-363) the parameters are pretty much explained or at least mentioned

Lines 382--390 practically run the Markov Chains and extract the posterior distributions of the parameters indicated in the "variable.names" parameter.
From that point on it is just a matter of choosing the right statistics (means, standard deviations and quantiles) of each quantity, assemble them into a big list (lines 436-458) and save everything.
Please if you have questions about the smaller details (i.e. the sub sections of the model code), let me know, I hope I can remember everything within a reasonable time... :)

In general the names of the variables and the quantities are the same that I have used in the paper, but each one should be specified in the commented parts.

3) **Plot code**
**1) Multi-panels** this is just a matter of retrieving what I have saved from the code above (lines 33-35, file newplots_Nov19.R).
Then the plots are done in groups: prevalence, screening, durations, incidence, completeness of reporting system and case reports.
For each one of these I have an external function in the folder containing all the plot parameters, actually plotting the results, plotting the data, plotting the splines smoothing the data, plotting Satterwhite reference data (for prevalence and incidence) and saving everything to keep the estimates.
Durations,screening, completeness of reporting and case reports had a prior and a posterior plot.
This may take a while to go through, but I don't know if it's crucial for your scope here. It's messy, I know.

**2) Plot for paper** is also longer but more linear (at least, it's only one code with no calls to external functions), and again is going through the results, extracting the ones I wanted, and plotting them with a nicer interface (ggplot). This also should contain the codes for the tables in the paper.

Again, take your time, and feel free to ask me anything you don't understand, I hope I am able to give you convincing answers. It's been a lot of work, and a lot of time since I have gone through this in detail.
There may be some bugs, yes. I hope not, but in such a long code it's almost certainty, as this has never been revised by anyone else than me.

Let me know!
Emanuele

On Thu, Sep 30, 2021 at 12:01 PM Roenn, Minttu Minna-Maarit <mronn@hsph.harvard.edu> wrote:

> Super! Thank you Emanuele! Much appreciated. -Minttu
>
> > On Sep 30, 2021, at 11:59 AM, Emanuele Mazzola <mazzola@ds.dfci.harvard.edu> wrote:
> >
> > Hi Minttu,
> > sure, no problem. I just have to look back myself because it's been quite a while since I ran it last time and I need to go back at what was the final version.
> > I'll get there.
> > Be aware, it's multiple files, but I'll explain to you the dependencies

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>
> Will get back to you soon.
> Emanuele
>
> On Thu, Sep 30, 2021 at 11:35 AM Roenn, Minttu Minna-Maarit <mronn@hsph.harvard.edu> wrote:
>> Hi Emanuele,
>>
>> I wanted to touch base with you on potentially getting access to your model code used in the paper. I would like to learn Stan, and Josh suggested I start by taking your code in rjags and move to Stan from there. There is also potential of trying to start a project with the CDC on state-level predictions for different STDs, which may leverage the national level framework you developed. All under works, and directions remain uncertain, but for me it would be educational to get to learn from an established and working model in any case.
>>
>> On the PPML github page, the last udpate is from June 2017, and I thought there might have been changes done since then. Would it be possible to push the latest version to github (or send it over)?
>>
>> Thanks in advance, and best of luck with the resubmission!
>>
>> All the best,
>> Minttu
>
> --
> ****************************************************************************
> **NOTE: please update your address book with THIS email address, because mazzola@jimmy.harvard.edu will be discontinued. Thanks!**
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